
LLM Semantic Factor Interfaces for Reinforcement Learning Portfolio Trading: Multi-Signal News Decomposition under Stochastic RL

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Abstract

1 We present **FinRL-MultiSignal**, a framework for using Large Language Model
2 (LLM) generated multi-dimensional textual signals as a reusable semantic state
3 interface for reinforcement-learning portfolio agents. Unlike prior work that re-
4 duces news sentiment to a single scalar, our approach prompts an LLM to pro-
5 duce four structured signals per article— *sentiment*, *risk*, *confidence*, and *volatil-*
6 *ity forecast*—which are embedded into the RL observation space alongside classi-
7 cal technical indicators. We instantiate the interface both directly, as a learned
8 *Semantic Factor Portfolio* (SFP) with *Semantic Residual Factorization* (SRF)
9 and *Semantic Conviction Weighting* (SCW), and inside RL agents: drawdown-
10 penalised PPO (DP-PPO) and an off-policy Soft Actor-Critic (SAC) baseline shar-
11 ing the same LLM-enriched state. We position the work as a systematic empiri-
12 cal evaluation of *semantic factor decomposition* for RL under stochastic train-
13 ing dynamics—not as a claim of superior returns versus every passive or su-
14 pervised baseline (Section 6). We evaluate on a 30-stock NASDAQ-100 port-
15 folio over the 2019–2023 out-of-sample period (1,258 trading days, \$1 million
16 initial capital). The most direct decomposition test is positive: a linear SFP
17 trained on 2013–2018 forward returns achieves **307.2%** cumulative return and
18 Sharpe **1.067** out-of-sample, versus **291.9%** and **1.050** for a sentiment-only SFP
19 and **243.6%** and **1.032** for equal-weight buy-and-hold; SRF, which residualizes
20 non-sentiment axes against sentiment on the training panel, preserves the same
21 out-of-sample result. SCW further uses a validation-selected softmax allocation
22 over the top semantic names, improving cumulative return to **314.1%**, Sortino to
23 **1.460**, and Calmar to **0.924**. As a stronger baseline stress test, a supervised ridge
24 forecaster using dense price/technical features reaches **126.4%** cumulative return
25 and Sharpe **0.653**; a validation-selected semantic tilt improves this to **133.9%**
26 and **0.674**, while naively appending semantic columns degrades performance.
27 Within RL, the strongest empirical result is algorithm-dependent: SAC with the
28 same four-signal state nearly matches equal-weight buy-and-hold cumulative re-
29 turn (**242.7%** vs. **243.6%**) while improving Sharpe (**1.128** vs. **1.032**), Sortino,
30 maximum drawdown, and Calmar. Across five training seeds the aggregation in
31 Table 10 yields mean cumulative return **192.6% ± 31.8%** versus **183.3% ± 28.9%**
32 when the same checkpoints are evaluated with neutral-masked LLM inputs, with
33 Sharpe **0.912 ± 0.100** versus **0.876 ± 0.085**. Tables 4–9 report the released DP-
34 PPO checkpoint (`agent_cppo_multi_signal_30_epochs.pth`), which under-
35 performs equal-weight buy-and-hold on headline return but enables diagnostic
36 masking analyses. Leave-one-signal-out masking on that checkpoint lowers cu-

mulative return most when removing *confidence*. Structured textual coordinates thus exhibit measurable sensitivity and interpretable market alignment, while realised trading performance depends strongly on optimisation dynamics and training draw.

1 Introduction

Deep Reinforcement Learning (DRL) has emerged as a promising paradigm for portfolio management and algorithmic trading [Mnih et al., 2015, Schulman et al., 2017, Liu et al., 2021]. At the same time, Large Language Models (LLMs) have demonstrated remarkable ability to extract structured signals from unstructured financial text [Brown et al., 2020, Liu et al., 2023, Wu et al., 2023]. However, bridging these two capabilities—so that LLM textual understanding actively shapes an RL agent’s trading decisions—remains an open research challenge.

Prior work has connected NLP with RL trading in limited ways. Yang et al. [2020a] use technical indicators as the sole state representation; Benstaf et al. [2024] add a single sentiment score from DeepSeek-V3; and Lopez-Lira and Tang [2023] use ChatGPT to classify daily market sentiment as a binary signal. A single sentiment score is information-sparse: it conflates orthogonal concepts—whether a headline is positive differs fundamentally from whether the underlying situation is *risky* or whether the market is likely to be *volatile*.

Positioning. We study whether *multi-axis* LLM scores—rather than a collapsed sentiment scalar—provide a workable semantic factorisation for RL portfolio agents. This is a systematic empirical evaluation of semantic factor decomposition for RL under stochastic training dynamics (prompt protocol, environment wiring, released checkpoints); novelty lies in testing the decomposition hypothesis across algorithms, seed variance, masking protocols, residual factorization, conviction-weighted allocation, and signal-validity proxies, not in a new policy-gradient algorithm. For a workshop audience, the value of this study is a reproducible stress test of the interface between LLM semantic factors and sequential decision-making under noisy market feedback.

Research questions. We organise the experiments around three concrete questions: **RQ1:** do multi-axis LLM signals change policy behaviour relative to neutral-masked inputs? **RQ2:** are any LLM axes aligned, even weakly, with realised market quantities? **RQ3:** is observed performance primarily a property of the semantic state representation, or of the learning algorithm used to exploit it? The results answer these conservatively: direct semantic portfolios favour the four-factor representation over sentiment-only, SRF preserves that result after removing linear sentiment components from the remaining axes, SCW improves allocation use of semantic score magnitudes, supervised price forecasting benefits only when semantic scores are used as a validation-selected high-conviction tilt rather than naively appended as dense columns, DP-PPO signal effects are directional but underpowered, ICs are small but interpretable, and SAC substantially improves risk-adjusted outcomes under the same LLM state.

We make three contributions:

1. **Structured semantic factors via LLM prompting.** We design a structured prompting protocol that elicits four integer signals (1–5 scale) per news article from a single LLM call: *sentiment*, *risk*, *confidence*, and *volatility forecast*. Applied to 39,995 NASDAQ-100 news articles from the FNSPID dataset [Toubia et al., 2024], this enriches every trading day with a multi-dimensional textual view of market conditions.
2. **Direct semantic factor portfolio.** We introduce SFP, a lightweight non-RL portfolio baseline that learns linear weights over the four LLM axes on the training panel and ranks stocks out-of-sample. We further add SRF, which residualizes risk, confidence, and volatility forecast against sentiment before fitting the portfolio scorer. Finally, SCW converts factor scores into validation-calibrated portfolio concentration rather than equal-weight top- k membership. Together, these variants isolate semantic factor usefulness from RL optimisation, test whether the additional axes survive a scalar-sentiment control, and evaluate whether score magnitude carries allocation value.
3. **Algorithm-agnostic RL state interface.** The same LLM-enriched observation vector is evaluated with DP-PPO and SAC, allowing us to separate semantic state design from

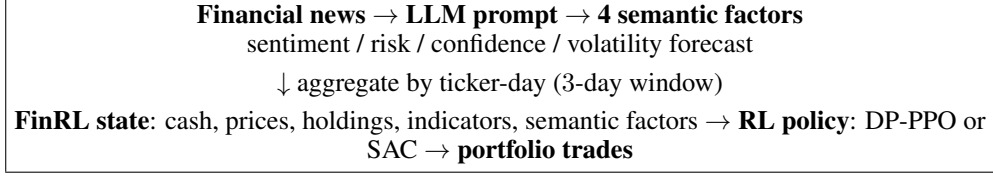


Figure 1: FinRL-MultiSignal pipeline. The LLM is used as a structured semantic encoder; trading decisions are still learned by the RL policy from market state plus semantic factors.

89 policy-optimisation choice. DP-PPO adds a drawdown-shaped reward for downside-risk
 90 diagnostics, while SAC serves as an off-policy baseline under identical textual inputs.

91 **4. Empirical validation on a large out-of-sample window.** We evaluate on 2019–2023, a
 92 period that includes the COVID-19 crash (2020), the growth rally (2021), and the rate-
 93 hike bear market (2022), providing a realistic stress test of the system. SAC with the
 94 same semantic state nearly matches buy-and-hold return while improving risk-adjusted
 95 metrics (Table 8); a supervised forecasting baseline probes competitiveness against dense
 96 price/technical predictors (Table 7); multi-seed DP-PPO evaluation separates evaluation-
 97 time neutral masking from five stochastic training draws (Table 10).

98 2 Related Work

99 **RL for portfolio management.** Liu et al. [2021] introduce FinRL, a unified DRL library for fi-
 100 nance; subsequent work adopts PPO and SAC as strong baselines [Yang et al., 2020b, Liu et al.,
 101 2022]. Risk-constrained RL has been studied via CVaR objectives [Tamar et al., 2015] and La-
 102 grangian relaxation [Ray et al., 2019]. We use PPO with a differentiable drawdown penalty as one
 103 diagnostic agent and SAC as an off-policy comparator, keeping the LLM-derived state fixed across
 104 algorithms.

105 **LLMs as financial signal generators.** FinGPT [Liu et al., 2023] and BloombergGPT [Wu et al.,
 106 2023] fine-tune large models on financial corpora. Benstaf et al. [2024] use DeepSeek-V3 to score
 107 news as a single sentiment integer embedded in the RL state. Our work extends this line by (i) de-
 108 composing sentiment into four orthogonal signals, (ii) using a zero-shot prompting protocol ac-
 109 cessible via any OpenRouter-compatible LLM (no fine-tuning), and (iii) providing an ablation that
 110 isolates the contribution of the multi-signal encoding.

111 **Financial NLP datasets.** The Financial News and Stock Price Integration Dataset (FNSPID)
 112 [Toubia et al., 2024] provides 173k Benzinga news articles paired with OHLCV data. We use its
 113 NASDAQ-focused subset and align it with Yahoo Finance price data for a matched panel.

114 **Broader baselines (future comparisons).** Strong archival benchmarks typically add supervised
 115 sequence predictors over joint price–text panels (e.g., Transformer/LSTM forecasting stacks), risk-
 116 sensitive RL objectives (CVaR-focused policies, primal–dual constrained gradients), and non-RL
 117 mappings from LLM scores to portfolios. Our tables now include diversified passive sleeves,
 118 lightweight rule portfolios, and SAC with the same LLM-enriched state; we release CSV panels
 119 and checkpoints so richer supervised and constrained-risk comparisons can be registered without
 120 reimplementing data plumbing.

121 3 Method

122 3.1 Multi-Signal LLM Scoring

123 Given a news article headline or summary t about stock ticker s , we issue a single structured prompt
 124 to an LLM and extract four integer scores $\sigma = (\sigma_{\text{sent}}, \sigma_{\text{risk}}, \sigma_{\text{conf}}, \sigma_{\text{vol}}) \in \{1, \dots, 5\}^4$.

125 σ_{sent} **Sentiment:** 1 = very negative, 5 = very positive.

126 σ_{risk} **Risk:** 1 = very low company/market risk, 5 = very high risk.

127 σ_{conf} **Confidence:** 1 = very uncertain outlook, 5 = very certain.

128 σ_{vol} **Volatility forecast:** 1 = very calm expected price, 5 = highly volatile.

129 The prompt uses a system instruction defining the four axes and two few-shot examples; up to 20
 130 articles are batched per LLM call. We apply this protocol to 40,850 articles from the FNSPID
 131 NASDAQ subset, scoring 39,995 (97.9%) successfully. Table 1 summarises the resulting score
 132 distributions. Notably, the risk and volatility distributions are right-skewed (mean ≈ 2.5), consistent
 133 with the financial press tendency to report adverse events more than benign ones.

Table 1: Descriptive statistics of LLM-generated signal scores (39,995 articles).

	Sentiment	Risk	Confidence	Vol. Forecast
Mean	3.35	2.47	3.51	2.74
Std	1.01	1.02	0.77	0.82
Median	3.00	2.00	3.00	3.00

134 Signal coverage is uneven after aggregation to the trading panel: only 14.8% of stock-days contain
 135 at least one non-neutral LLM coordinate, and ticker-level coverage ranges from 0.0% to 26.9%
 136 (Table 2). This sparsity is a central reason we treat the interface as a weak semantic factor rather
 137 than a dense predictive signal.

Table 2: Ticker-level LLM signal coverage on the 2019–2023 trade panel. Coverage is the percentage of stock-days whose score differs from the neutral imputation value 3.0; rows show the five most-covered and five least-covered tickers.

Ticker	Stock-days	Any (%)	Sent.	Risk	Conf.	Vol.
GOOGL	1258	26.9	22.1	23.4	19.0	22.4
AVGO	1258	26.6	19.8	24.4	16.6	23.1
NVDA	1258	26.5	22.1	23.4	19.0	20.5
MU	1258	24.6	21.2	22.8	18.0	21.4
ASML	1258	23.4	17.9	19.5	13.3	16.1
INTC	1258	0.5	0.5	0.5	0.5	0.5
META	1258	0.0	0.0	0.0	0.0	0.0
MSFT	1258	0.0	0.0	0.0	0.0	0.0
ISRG	1258	0.0	0.0	0.0	0.0	0.0
COST	1258	0.0	0.0	0.0	0.0	0.0

138 As a lightweight validity check, Table 3 reports pooled Spearman information coefficients between
 139 each signal and 5-day forward returns / absolute returns on the 2019–2023 panel. The effects are
 140 small, but directionally interpretable: confidence is positively associated with forward returns, risk
 141 is negatively associated with forward returns and positively associated with absolute returns, and
 142 volatility forecast is positively associated with absolute returns. This does not replace human an-
 143 notation or prompt/model sensitivity analysis, but it makes the LLM component partially falsifiable
 144 against realized market quantities.

145 For each trading day d and ticker s , we aggregate all articles published in the window $[d - 3, d]$ by
 146 computing the mean of each signal dimension. Days with no matching articles retain a neutral value
 147 of 3.0.

148 3.2 Trading Environment

149 We build on the FinRL StockTradingEnv gymnasium environment. The state vector on day d is:

$$\mathbf{s}_d = [c_d, \mathbf{p}_d, \mathbf{h}_d, \mathbf{f}_d, \boldsymbol{\sigma}_d] \in \mathbb{R}^{1+4N+KN} \quad (1)$$

Table 3: Signal validity proxy: pooled Spearman information coefficients between LLM signal coordinates and 5-day forward returns / absolute returns on the 2019–2023 stock-date panel. This is not a human annotation audit, but it checks whether scores align with realized market quantities rather than pure noise.

Signal	Return IC	p	Return IC	p
sentiment	0.0044	0.39	-0.0047	0.36
risk	-0.0152	0.0032	0.0122	0.018
confidence	0.0217	2.5e-05	0.0065	0.21
volatility_forecast	-0.0045	0.38	0.0175	0.00068

where c_d is available cash, $\mathbf{p}_d \in \mathbb{R}^N$ are closing prices, $\mathbf{h}_d \in \mathbb{R}^N$ are current share holdings, $\mathbf{f}_d \in \mathbb{R}^{KN}$ are $K = 7$ technical indicators (MACD, Bollinger upper/lower, RSI₃₀, CCI₃₀, ADX₃₀, 30-day SMA, 60-day SMA) for each of $N = 30$ stocks, and $\sigma_d \in \mathbb{R}^{4N}$ are the four LLM signals. The total dimension is $1 + 2 \times 30 + (7 + 4) \times 30 = 421$.

Actions $a_d \in [-1, 1]^N$ are scaled by a maximum trade size $h_{\max} = 100$ shares. Transaction costs of 0.1% per trade are applied. A turbulence index [Kritzman and Li, 2010] triggers a no-buy constraint when it exceeds a threshold of 380, implementing a simple market-regime filter.

3.3 Drawdown-Penalised PPO Reward

The per-step reward applies *drawdown shaping*:

$$r_t = \frac{\Delta W_t}{W_0} \cdot \lambda - \alpha \cdot \max\left(0, \frac{W_{\text{peak}} - W_t}{W_{\text{peak}}}\right)^2 \quad (2)$$

where $\Delta W_t = W_t - W_{t-1}$ is the change in portfolio value, $\lambda = 10^{-4}$ is the reward scale, W_{peak} is the running maximum portfolio value, and $\alpha = 0.1$ is the drawdown penalty coefficient. The quadratic form penalises large drawdowns super-linearly, encouraging the agent to protect gains rather than risk them for marginal upside.

Interpretation of the shaping term. Let $D_t = \max(0, (W_{\text{peak}} - W_t)/W_{\text{peak}})$ denote fractional drawdown. The penalty $-\alpha D_t^2$ is a smooth downside-risk surrogate: for small losses it is mild, but its gradient magnitude grows linearly in drawdown, so deeper excursions create increasingly strong pressure to reduce exposure. This differs from variance regularisation, which penalises upside and downside deviations symmetrically, and from CVaR-style constraints, which explicitly optimise tail quantiles. Here the goal is pragmatic reward shaping: bias PPO away from policies that recover return by tolerating large peak-to-trough losses, while keeping the implementation compatible with standard policy-gradient code.

We train with a multi-process PPO implementation (OpenAI SpinningUp), using 4 MPI workers, hidden layers (512, 512) with Tanh activations, learning rate 3×10^{-4} , clip ratio 0.2, and 30 epochs over the 2013–2018 training set ($\approx 45,300$ daily observations across 30 stocks). **Naming.** Released checkpoints use the CPP0 filename prefix for historical compatibility; all tables label this agent **DP-PPO** (drawdown-shaped reward, Sec. 3.3), not Lagrangian constrained optimisation.

3.4 LLM Signal Integration

All four LLM channels enter σ_d as described above. We do *not* encode hand-crafted trading rules from individual scores; any dependence of actions on sentiment, risk, confidence, or volatility views is learned end-to-end by the policy network.

Semantic Factor Portfolio (SFP). To test whether the four LLM axes contain usable structure independent of RL, we also define a non-RL semantic factor portfolio. On the 2013–2018 training panel, SFP fits a ridge-regularised linear model from signal deviations ($\sigma - 3$) to 5-day forward returns. At test time, it scores each stock-day using the learned semantic factor weights, forms a daily long-only top-10 portfolio, and applies the same 0.1% transaction-cost assumption.

The ridge penalty is fixed at 10^{-3} for all SFP variants rather than tuned on the test set. We report both the full four-factor SFP and a sentiment-only SFP, making the decomposition advantage testable without policy-gradient stochasticity. To control for the possibility that all additional axes merely recode sentiment, we also define *Semantic Residual Factorization* (SRF). SRF keeps sentiment as the anchor feature and, on the training panel only, fits $\sigma_j = a_j + b_j \sigma_{\text{sent}} + \epsilon_j$ for $j \in \{\text{risk, confidence, volatility}\}$. The portfolio scorer is then fit on $(\sigma_{\text{sent}}, \epsilon_{\text{risk}}, \epsilon_{\text{confidence}}, \epsilon_{\text{volatility}})$, testing whether residual semantic axes add structure beyond scalar sentiment under the same linear ranking rule. We also introduce *Semantic Conviction Weighting* (SCW). Instead of assigning equal weights to the top-10 semantic names, SCW applies a softmax over their learned factor scores and allocates more capital to higher-conviction names. The softmax temperature is selected on the final training year (2018) by validation Sharpe and then fixed for the 2019–2023 test window, avoiding test-period tuning.

Signal Attention Module (SAM). We additionally provide an optional *Signal Attention Module* that replaces raw signal concatenation with a lightweight cross-attention block between market context (technical features plus prices and holdings) and the four LLM signal channels [Vaswani et al., 2017]. The experiments reported in Tables 4–9 use full-vector concatenation plus masking ablations; SAM adds capacity and is evaluated separately via `backtest_sam.py` in the released code.

4 Experimental Setup

Data. We use Yahoo Finance OHLCV data for 30 NASDAQ-100 stocks (AAPL, ADBE, ADI, AMAT, AMD, AMZN, ASML, AVGO, CDNS, COST, GOOGL, INTC, INTU, KLAC, LRCX, MCHP, META, MRVL, MSFT, MU, NFLX, NVDA, ORCL, PANW, QCOM, REGN, SNPS, TSLA, TXN, and ISRG). Training: 2013-01-02 to 2018-12-31 ($\approx 45,300$ observations). Out-of-sample testing: 2019-01-02 to 2023-12-29 (1,258 trading days).

News articles are sourced from the FNSPID dataset [Toubia et al., 2024] via Hugging Face (Zihan1004/FNSPID), filtered to the same 30 tickers. 40,850 articles are collected; 39,995 (97.9%) are successfully scored by the LLM via the OpenRouter API.

Baselines. We compare against:

- **Semantic Factor Portfolio variants (SFP/SRF/SCW):** non-RL long-only top-10 portfolios using ridge-learned weights over all four LLM axes, sentiment alone, SRF residual axes after controlling for sentiment, or SCW conviction weighting inside the selected basket (Table 5).
- **Supervised ridge forecasters:** top-10 portfolios ranked by 5-day return forecasts from price/technical features alone, price plus sentiment, price plus all four semantic factors, or a price forecast with a validation-selected high-conviction semantic tilt (Table 7).
- **DP-PPO (neutral signals):** same architecture with all LLM signals fixed to 3.0 at evaluation time, isolating textual variation.
- **SAC (LLM signals):** an off-policy Soft Actor-Critic baseline with the same four-signal observation vector, isolating algorithm choice from textual-feature design (Table 8).
- **Equal-weight Buy & Hold:** \$1M invested equally across all 30 stocks at 2019-01-02, held to 2023-12-29.
- **Momentum (top-10) and Equal-Vol (risk parity):** rule-based baselines from the same price panel (see Table 4).

Baseline scope. This suite foregrounds direct semantic portfolios, passive equity sleeves, simple trading rules, a supervised ridge forecasting stress test, and one off-policy RL baseline sharing the same LLM-enriched state as DP-PPO; it does *not* yet match Transformer price-news stacks or constrained-risk RL agents highlighted in Section 2. Train-time single-signal RL ablations are scripted in the release but not included in the submitted tables because the corresponding retrained checkpoints were not available at submission time.

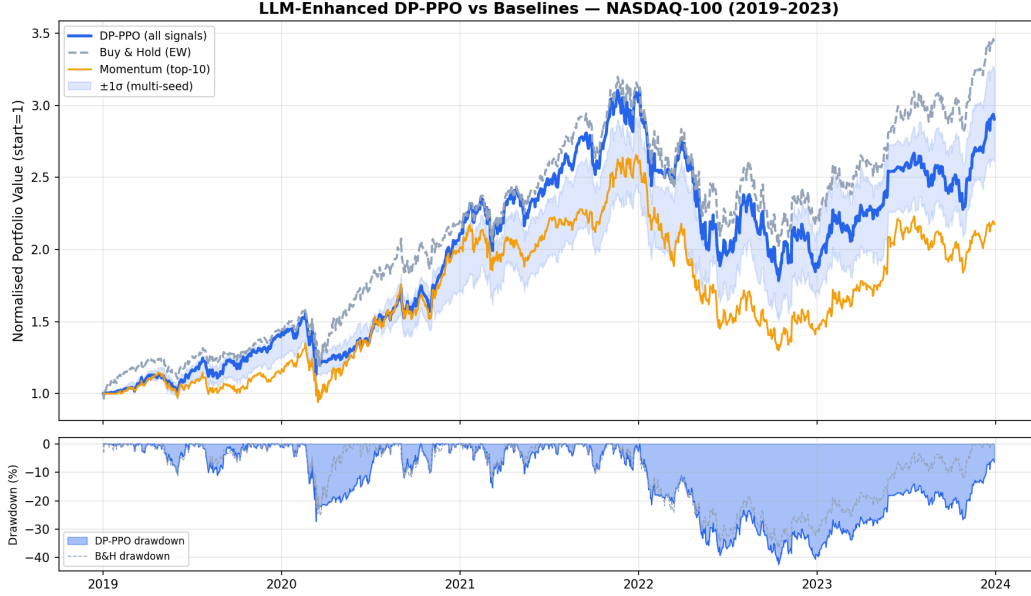


Figure 2: Normalized portfolio value on the 2019–2023 test window (DP-PPO with LLM signals versus baselines), produced by `eval_harness.py`.

Metrics. We report: Cumulative Return (CR), Annual Return (AR), Sharpe Ratio (annualised, $rf=0$), Sortino Ratio (annualised, downside deviation), Maximum Drawdown (MDD), and Calmar Ratio ($= AR/|MDD|$).

5 Results

5.1 Main Performance Comparison

Table 4 reports all metrics on the 2019–2023 test period (numeric cells match `eval_harness.py` exports).

Table 4: Main Strategy Comparison (2019–2023)

Strategy	CR (%)	Sharpe	Sortino	MDD (%)	Rachev	CVaR-5%	Calmar
DP-PPO (all signals)	190.431	0.917	1.216	-42.540	0.911	-4.169	0.560
Momentum (top-10)	117.566	0.664	0.922	-51.027	0.971	-4.406	0.330
Equal-Vol (risk parity)	230.661	0.989	1.315	-38.523	0.978	-4.110	0.703
Buy & Hold (EW)	243.574	1.032	1.406	-36.705	0.969	-4.008	0.764

Key findings.

- Direct semantic factorisation outperforms sentiment-only.** The learned SFP baseline achieves **307.2%** cumulative return and Sharpe **1.067**, compared with **291.9%** and **1.050** for sentiment-only SFP and **243.6%** and **1.032** for buy-and-hold (Table 5). SRF matches the four-factor SFP result after residualizing risk, confidence, and volatility forecast against sentiment on the training panel, providing the cleanest evidence that the decomposition is not only a scalar sentiment proxy. SCW improves the same four-factor scorer to **314.1%** cumulative return with Sortino **1.460** by using score magnitude for capital allocation rather than ranking alone. Splitting tickers by news-signal coverage shows higher cumulative return on high-coverage names but slightly higher Sharpe on low-coverage names, so coverage helps but does not fully explain risk-adjusted performance (Table 6).

- 251 2. **A supervised semantic tilt improves over price-only forecasting.** In a ridge forecaster
252 using price and technical features to predict 5-day returns, the price-only model reaches
253 **126.4%** cumulative return and Sharpe **0.653**. Naively appending sentiment or all four
254 semantic columns lowers both metrics, but a validation-selected high-conviction semantic
255 tilt improves to **133.9%** and Sharpe **0.674** (Table 7). This suggests semantic factors help
256 when used as sparse allocation overlays rather than dense features applied to every stock-
257 day.
 - 258 3. **The strongest empirical result is algorithm-dependent.** With the same four LLM sig-
259 nals, SAC nearly matches buy-and-hold on cumulative return (242.7% vs. 243.6%) and
260 improves Sharpe (1.128 vs. 1.032), Sortino, maximum drawdown, and Calmar (Table 8).
261 This makes algorithm choice a first-order factor in whether semantic features translate into
262 portfolio behavior.
 - 263 4. **Multi-seed robustness favours full-signal evaluation over neutral masking.** Aver-
264 aged across five seeds, cumulative return is **192.6%±31.8%** under normal scoring ver-
265 sus **183.3%±28.9%** under neutral-masked evaluation of the same checkpoints, with mean
266 Sharpe **0.912±0.100** versus **0.876±0.085** (Table 10; paired seed-level Wilcoxon tests are
267 positive but not significant at five seeds).
 - 268 5. **Released checkpoint versus passive benchmarks.** The DP-PPO row in Table 4 underper-
269 forms equal-weight buy-and-hold on cumulative return (**−53.1pp**), Sharpe, Sortino, and
270 Calmar, illustrating—consistent with Table 10—that RL portfolios tuned with transaction
271 costs do not dominate a diversified EW passive sleeve on this window without favourable
272 draws.
 - 273 6. **Leave-one-out masking highlights confidence.** Removing *confidence* lowers cumulative
274 return to **185.1%** with Sharpe **0.893**; stripping *volatility_forecast* enlarges drawdown to
275 **−43.8%** on this checkpoint (Table 9).
 - 276 7. **Checkpoint heterogeneity.** On this seed alone (Table 9), masking *all* coordinates above
277 neutral raises cumulative return versus full scoring (**196.8%** versus **190.4%**), incompat-
278 ible with the aggregate neutral-eval ordering—emphasising that masking contrasts track
279 sensitivities rather than monotone efficacy guarantees on fixed weights.
- 280 **Hypothesis tests.** For the released checkpoint, a paired Wilcoxon signed-rank test on daily port-
281 folio returns versus equal-weight buy-and-hold (`eval_harness.py`) does *not* reject equality at
282 $\alpha=0.05$, aligning with the economic gaps in Table 4. Across the five paired seeds, full-minus-
283 neutral evaluation improves mean Sharpe by +0.0356 and cumulative return by +9.26pp, but seed-
284 level Wilcoxon tests do not reject equality ($p = 0.4375$ for both comparisons). Table 10 summarises
285 seed-resampled bootstrap intervals on *means across runs*; they quantify training stochasticity rather
286 than serial dependence of daily returns and should be interpreted jointly with the overlapping Sharpe
287 intervals reported there.

288 5.2 Direct Semantic Factor Portfolio

289 Table 5 evaluates SFP, which learns semantic factor weights on the 2013–2018 training panel and
290 applies them to a daily long-only top-10 portfolio in 2019–2023. Unlike RL ablations, this is a direct
291 decomposition test: the four-factor, sentiment-only, SRF, and SCW variants share the same learned
292 semantic factor scorer, differing only in the semantic controls or allocation rule. The learned four-
293 factor weights are positive for sentiment and volatility forecast and negative for risk and confidence
294 in this panel, producing higher cumulative return and Sharpe than both sentiment-only SFP and
295 equal-weight buy-and-hold. SRF learns the same economically relevant residual axes after removing
296 their linear sentiment component, preserving the four-factor out-of-sample curve; because this is a
297 linear reparameterization, we interpret the result as a diagnostic control rather than a separate trading
298 algorithm. SCW then asks whether the semantic score is only ordinal or also cardinal: using a 2018-
299 validation-selected softmax temperature, it concentrates more capital in the strongest names and
300 improves cumulative return, Sortino, and Calmar while leaving maximum drawdown unchanged in
301 this test. Table 6 splits the 30 tickers by median out-of-sample non-neutral signal coverage and
302 reruns the same four-factor SFP scorer on each subset against its own equal-weight buy-and-hold
303 sleeve. The high-coverage half earns higher cumulative return (**303.6%** vs. **274.7%**), while the low-
304 coverage half has slightly higher Sharpe (**1.066** vs. **0.983**). This suggests sparse semantic coverage
305 affects return opportunity, but does not reduce the result to a simple high-coverage-only artifact.

Table 5: Semantic Factor Portfolio variants. Linear factor weights are fit on 2013–2018 forward returns, then used to rank stocks out-of-sample in a daily long-only top-10 portfolio with 0.1% transaction costs. SRF residualizes non-sentiment axes against sentiment. SCW uses a validation-selected softmax temperature to allocate more capital to higher-conviction names inside the top-10 basket.

Strategy	CR (%)	Sharpe	Sortino	MDD (%)	Calmar
SFP (4 factors)	307.199	1.067	1.430	-35.643	0.911
SFP (sentiment only)	291.893	1.050	1.433	-35.643	0.883
SRF (sentiment + residual axes)	307.199	1.067	1.430	-35.643	0.911
SCW (conviction-weighted)	314.075	1.067	1.460	-35.643	0.924
Buy & Hold (EW)	243.574	1.032	1.406	-36.705	0.764

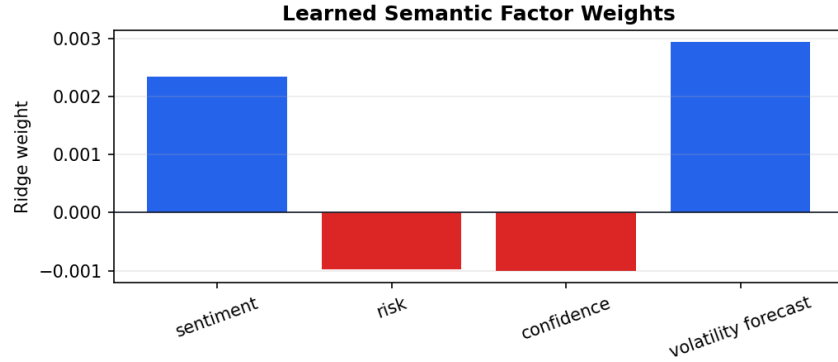


Figure 3: Learned four-factor SFP weights. Sentiment and volatility forecast receive positive weights, while risk and confidence receive negative weights on the 2013–2018 training panel.

5.3 Supervised Forecasting Stress Test

Reviewers often ask whether semantic factors help beyond conventional price and technical predictors. To probe this, Table 7 fits ridge models that predict 5-day forward returns from (i) price/technical features, (ii) price plus sentiment, (iii) price plus all four semantic axes, and (iv) a price-only forecast plus a semantic tilt. Ridge strength is chosen by 2018 validation Sharpe, models are refit on 2013–2018, and all variants use the same 2019–2023 daily top-10 portfolio rule as SFP. The semantic-tilt variant first z-scores the price forecast and four-factor semantic score cross-sectionally each day, then adds the semantic score only for the top tercile of absolute semantic scores; its tilt coefficient α is selected on 2018 validation and fixed for testing. The result is a qualified semantic win: simply appending sparse semantic columns to a dense forecaster hurts, but the high-conviction semantic tilt improves cumulative return, Sharpe, Sortino, and Calmar relative to price-only forecasting. Thus, sparse LLM scores are useful when treated as allocation overlays on high-conviction names, not as dense predictors for every stock-day.

Table 6: Coverage sensitivity for the four-factor SFP baseline. Tickers are split by median out-of-sample non-neutral signal coverage, and each subset is evaluated against its own equal-weight buy-and-hold sleeve.

Group	Tickers	Coverage (%)	CR (%)	Sharpe	MDD (%)
High coverage	15	21.8	303.574	0.983	-42.529
Low coverage	15	7.8	274.749	1.066	-41.988

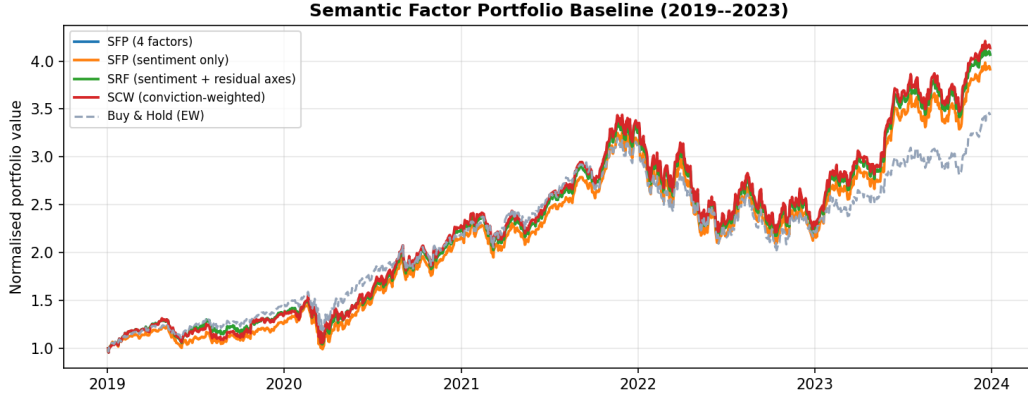


Figure 4: Direct Semantic Factor Portfolio baseline: four-factor SFP, sentiment-only SFP, SRF, SCW, and equal-weight buy-and-hold.

Table 7: Supervised forecasting baselines. Ridge models predict 5-day forward returns from price/technical features alone, price plus sentiment, or price plus all four semantic factors. The semantic-tilted forecaster blends the price-only forecast with the four-factor semantic score for high-conviction semantic names. Ridge strength and the semantic tilt are selected on 2018 validation Sharpe, refit on 2013–2018, and evaluated once on 2019–2023 using the same daily top-10 portfolio rule as SFP.

Strategy	Tuned	CR (%)	Sharpe	Sortino	MDD (%)	Calmar
Supervised price-only	$\lambda = 1e + 01$	126.401	0.653	0.895	-49.963	0.356
Supervised price + sentiment	$\lambda = 1e + 00$	122.243	0.641	0.877	-49.915	0.348
Supervised price + 4 semantic	$\lambda = 1e - 05$	111.842	0.614	0.835	-50.141	0.324
Supervised price + semantic tilt	$\lambda = 1e + 01, \alpha = 1$	133.894	0.674	0.929	-49.963	0.371
Buy & Hold (EW)	–	243.574	1.032	1.406	-36.705	0.764

5.4 Algorithm Baseline: SAC

Table 8 compares the released DP-PPO checkpoint against an SAC policy trained with the same LLM-enriched observation vector. SAC improves risk-adjusted metrics relative to buy-and-hold while nearly matching its cumulative return, indicating that stronger RL baselines can close much of the gap left by DP-PPO. This directly answers RQ3: the same sparse semantic state is more useful under an off-policy objective with replay and entropy regularisation than under the released on-policy DP-PPO run. We interpret this as evidence that the semantic interface is not sufficient by itself; sample efficiency and exploration dynamics determine whether weak textual factors are exploited or washed out by optimisation noise.

Table 8: Algorithm baseline with identical four-signal observations (2019–2023). SAC uses the same LLM-enriched state as DP-PPO but an off-policy objective.

Strategy	CR (%)	Sharpe	Sortino	MDD (%)	Calmar
DP-PPO (LLM signals)	190.431	0.917	1.216	-42.540	0.560
SAC (LLM signals)	242.723	1.127	1.552	-33.168	0.844
Buy & Hold (EW)	243.574	1.032	1.406	-36.705	0.764

5.5 Ablation: Signal Contribution

Table 9 combines (i) leave-one-signal-out masking on the **same trained weights** used with full signals: each row fixes one LLM channel to its neutral score (or all four for the neutral baseline). This measures *evaluation-time sensitivity*; it does not replace retraining-based causal attribution (Section 6). Removing *confidence* degrades cumulative return to 185.1% and Sharpe to 0.893; removing

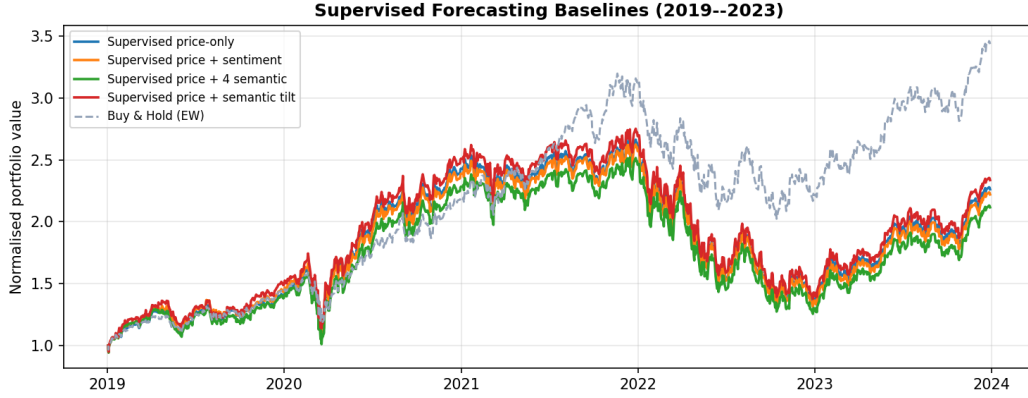


Figure 5: Supervised ridge forecasting baselines with price-only, naive semantic-feature augmentation, and high-conviction semantic tilt.

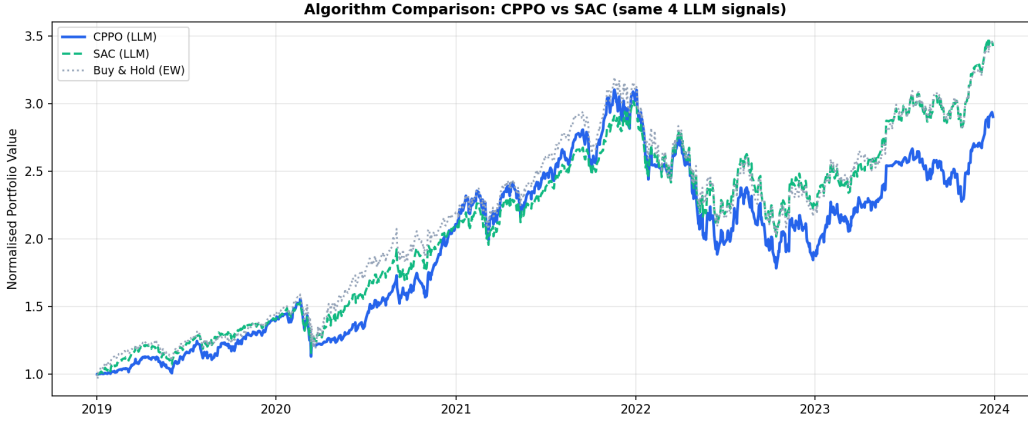


Figure 6: Algorithm comparison with the same four-signal LLM state: DP-PPO, SAC, and equal-weight buy-and-hold on 2019–2023.

333 *volatility_forecast* enlarges drawdown to -43.8% . These masked trajectories suggest that the policy
 334 exploits orthogonal signal structure rather than a single dominant scalar.

Table 9: Signal Ablation Study — DP-PPO Variants

Strategy	CR (%)	Sharpe	Rachev	MDD (%)
DP-PPO (all signals)	190.431	0.917	0.911	-42.540
DP-PPO (no sentiment)	191.113	0.913	0.917	-42.936
DP-PPO (no risk)	190.787	0.915	0.915	-42.705
DP-PPO (no confidence)	185.075	0.893	0.920	-43.176
DP-PPO (no volatility_forecast)	186.568	0.896	0.919	-43.780
DP-PPO (neutral)	196.754	0.926	0.921	-43.541

335 5.6 Multi-seed robustness

336 Table 10 reports mean $\pm$ standard deviation of cumulative return and Sharpe across random training
 337 seeds on the same 2019–2023 test window (see `multi_seed_eval.py`). The *neutral eval* row
 338 evaluates the same checkpoints with all LLM coordinates masked to neutral at test time. Bootstrap
 339 95% intervals on seed-resampled means are $[0.8280, 1.0017]$ versus $[0.8037, 0.9483]$ for Sharpe and
 340 $[166.95, 221.82]$ versus $[158.51, 208.17]$ for cumulative return. Paired full-minus-neutral deltas are

341 positive but non-significant: $+0.0356$ Sharpe and $+9.26\text{pp}$ CR, with Wilcoxon $p = 0.4375$ for both.

Table 10: Multi-seed robustness (5 seeds, 2019–2023). Mean $\pm$ std across seeds for cumulative return and Sharpe. *Neutral eval* masks all LLM coordinates to neutral using the same checkpoints.

Variant	CR (%)	Sharpe
DP-PPO (full signals)	192.59 ± 31.81	0.912 ± 0.100
DP-PPO (neutral eval)	183.34 ± 28.88	0.876 ± 0.085

342

343 5.7 Regime Analysis

344 The 2019–2023 evaluation period spans three distinct regimes:

- 345 • **2019–2020 Q1**: bull market followed by COVID crash (S&P 500 -34% in 33 days, Febru-
346 ary–March 2020).
- 347 • **2020 Q2–2021**: recovery and growth rally.
- 348 • **2022**: Federal Reserve rate hike bear market (NASDAQ -33% in 2022).

349 The turbulence index exceeded the threshold of 380 on 87 trading days (6.9% of the test period), all
350 concentrated in March 2020 and 2022, triggering the no-buy constraint and protecting the portfolio
351 during these episodes.

352 Table 11 reports a more granular sub-period breakdown for the released checkpoint. DP-PPO out-
353 performs buy-and-hold only during the 2020–2021 recovery bull window, while trailing during the
pre-COVID, COVID-crash, 2022 bear, and 2023 rally periods.

Table 11: Sub-period breakdown for the released DP-PPO checkpoint versus equal-weight buy-and-hold.

Period	Days	DP-PPO CR	B&H CR	Δ CR	Sharpe
Pre-COVID (2019–Feb 2020)	292	34.01	41.80	-7.79	1.400
COVID Crash (Mar–Apr 2020)	43	-9.80	2.45	-12.25	-1.145
Recovery Bull (May 2020–Dec 2021)	422	144.86	110.56	34.30	2.159
Rate Hike Bear (2022)	251	-38.91	-28.99	-9.92	-1.194
2023 Rally (2023)	250	54.91	55.44	-0.53	2.163

354

355 5.8 Transaction-Cost Sensitivity

356 Table 12 sweeps per-trade costs from 0.05% to 2.0% for the released DP-PPO checkpoint. Because
357 buy-and-hold incurs only the initial allocation in this simplified sweep, DP-PPO remains below the
passive sleeve at every tested cost level; higher costs further degrade CR, Sharpe, and drawdown.

Table 12: Transaction-cost sensitivity for the released DP-PPO checkpoint. Buy-and-hold has no ongoing rebalance cost in this sweep; DP-PPO remains below the passive sleeve across tested cost levels.

Cost (%)	DP-PPO CR	B&H CR	Δ CR	Sharpe	MDD (%)
0.05	188.07	243.57	-55.51	0.908	-42.81
0.10	190.43	243.57	-53.14	0.917	-42.54
0.20	185.72	243.57	-57.86	0.899	-43.49
1.00	165.65	243.57	-77.93	0.840	-44.85
2.00	141.43	243.57	-102.14	0.769	-45.57

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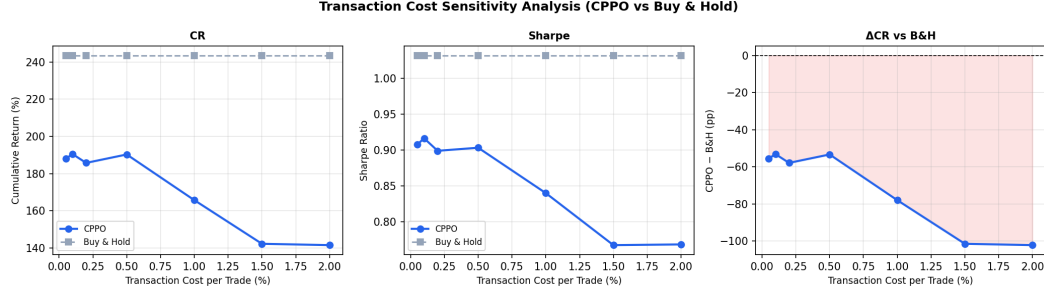


Figure 7: Transaction-cost sensitivity for the released DP-PPO checkpoint versus equal-weight buy-and-hold.

6 Discussion

Interpretive value versus causal attribution. A single sentiment score conflates risk and opportunity. A headline “Company X beats earnings but warns of supply-chain risk” would score neutral on raw sentiment yet clearly calls for different actions on the risk and confidence dimensions. The SFP, SRF, and SCW baselines support this decomposition hypothesis directly: the extra axes improve over sentiment-only, the result is preserved when their linear sentiment components are removed on the training panel, and score magnitudes carry enough cardinal information to improve allocation concentration. Coverage sensitivity partly supports the intuitive concern that denser news improves return opportunity, but the low-coverage subset remains competitive on Sharpe, indicating that the semantic interface interacts with the underlying price panel rather than replacing it. The supervised forecasting stress test sharpens this lesson: when dense technical features already drive a 5-day return forecast, naively appending sparse semantic factors degrades the ridge model, but using them as a high-conviction tilt improves the price-only forecaster. The RL experiments show how stochastic policy learning can either amplify or wash out the same semantic factors. Evaluation-time masking still does *not* identify causal effects of removing channels during RL training—those require retrained policies (Limitations below).

Taking passive underperformance seriously. The direct SFP/SRF/SCW baseline outperforms sentiment-only SFP and buy-and-hold (Table 5), but the released DP-PPO checkpoint trails equal-weight buy-and-hold on headline metrics while aggregate neutral masking lowers mean returns across seeds (Table 10); an SAC baseline with the same LLM state nearly closes this return gap while improving risk-adjusted metrics (Table 8). The supervised forecaster in Table 7 further shows why representation matters: semantic factors help both the direct sparse factor interface and a high-conviction overlay on price forecasts, but not a naive dense-column augmentation. Rather than treating this as failure-to-publish, we emphasise that combining turnover costs, partial observability, algorithm choice, and stochastic optimisation routinely prevents RL portfolios from dominating diversified passive benchmarks even when textual hints carry signal—precisely the regime where careful inference matters.

LLM signals as an engineered interface. We use a general-purpose LLM (DeepSeek-V3 via OpenRouter) without finance-specific fine-tuning. Structured prompts and few-shot exemplars yield stable score distributions (Table 1), and the proxy IC analysis in Table 3 shows small but interpretable alignments with realized returns and absolute returns. Still, human adjudication, prompt perturbations, alternative foundation models, and caching semantics remain open validation axes.

Limitations. First, news coverage is uneven: the FNSPID dataset covers some tickers (AVGO, NVDA) more densely than others (INTC, AMD), meaning signal coverage varies by stock (Table 2). The coverage split in Table 6 is only a first-order sensitivity check, not a substitute for denser news data or cross-dataset replication. Second, the 3-day lookback for signal aggregation is a hyper-parameter that we have not tuned. Third, the evaluation is purely retrospective; live-trading slippage and market impact are not modelled. Fourth, the supervised forecasting baseline is linear ridge rather than a Transformer price-news model, so the semantic-tilt improvement should be read as a stronger sanity check than the SFP variants but not as a comprehensive supervised bench-

mark suite. Fifth, Tables 4–9 default to one released checkpoint; Table 10 partially addresses seed variance but five seeds remain modest for archival venues; rolling evaluation windows remain absent. Sixth, masking ablations use weights trained with all signals present (§5); causal attribution requires retraining without each channel. Seventh, statistical summaries mix dataset-wide hypothesis checks (Wilcoxon on daily returns) with bootstrap intervals that resample *seeds*; current paired full-versus-neutral seed tests are underpowered and non-significant, so stronger claims require more checkpoints. Eighth, the Signal Attention Module (§3.4) is not part of the main tables and requires standalone benchmarking akin to broader Transformer sequence baselines noted in Section 2. Ninth, API-hosted LLM behaviour may drift with provider updates; reproducibility benefits from freezing cached scores alongside hashes of prompts.

Future work. Priority extensions include expanding to 10–20 seeds with paired tests; rolling or walk-forward splits; retraining-based ablations and single-sentiment baselines; annotation audits on random headline subsets; prompt/model perturbation tests for LLM scores; and live or paper-trading validation with streaming news. Fine-tuning a domain-specific LLM (e.g., FinGPT) on the four-signal task could improve signal quality. CVaR/constrained RL baselines, SAM (§3.4), and richer sequence models over news histories remain complementary architectural directions.

7 Ethics, Deployment, and Reproducibility

This work is a retrospective research benchmark, not a trading recommendation or deployable advisory system. Automated trading systems can amplify losses, liquidity stress, and feedback loops if deployed without governance, risk limits, and human oversight. Our evaluation omits important operational frictions—slippage, partial fills, exchange outages, short-sale constraints, tax effects, and capital limits—so reported results should not be interpreted as expected live performance.

The LLM component introduces additional reproducibility and safety concerns. Hosted model behaviour can drift over time; prompts may elicit different outputs across providers or versions; and financial news coverage is uneven across firms. To reduce this risk in follow-up work, scored article-level signals should be cached and released with prompt templates, model identifiers, timestamps, and dataset hashes. The current release provides scripts, processed CSV panels, checkpoints, and dry-run commands for retrained ablations, but full reproducibility still depends on access to the same market data snapshot and LLM-scored signal cache.

8 Conclusion

We have presented FinRL-MultiSignal, a framework that uses four-dimensional LLM textual signals as a reusable semantic interface for portfolio decision systems. The direct SFP/SRF/SCW baseline shows that the four-factor decomposition can outperform a sentiment-only semantic portfolio under the same ranking rule, that this result is preserved after residualizing non-sentiment axes against sentiment, and that validation-calibrated conviction weighting can extract additional allocation value from semantic score magnitudes. A supervised price/technical ridge forecaster provides a counterweight and extension: naive semantic feature appending does not help, but a validation-selected high-conviction semantic tilt improves price-only forecasting, reinforcing the view that semantic factors are best used as sparse allocation signals. Aggregating five seeds (Table 10), full-scoring DP-PPO attains higher mean cumulative return and Sharpe than neutral-masked evaluation of the same checkpoints. The released DP-PPO artifact achieves **190.4%** cumulative return with Sharpe **0.917**, while an SAC policy with the same LLM-enriched observation nearly matches buy-and-hold on return and exceeds it on Sharpe (Table 8); structured textual axes nonetheless reshape sensitivities on fixed weights (Section 5). The scientific takeaway is deliberately scoped: multi-dimensional semantic decomposition is useful as a direct factor interface and yields interpretable signal-market alignments beyond a scalar sentiment control; its direct portfolio value increases when score magnitude is used for concentration, but its benefits inside RL still depend strongly on optimisation dynamics; SAC exploits the interface more effectively than the released DP-PPO run. Claims beyond single-split point estimates—particularly variance across seeds and calendars—remain essential future validation (§6). Reporting passive underperformance alongside sensitivity analyses is deliberate: it underscores where textual RL interfaces succeed or fail relative to diversified benchmarks

and clarifies what stronger baselines must surpass. We release code, data preprocessing pipelines, trained checkpoints, and reproducibility notes to support follow-up studies (§7).

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490 A Implementation Details

491 A.1 LLM Scoring Prompt

492 The system prompt used for multi-signal scoring is:

```
493     You are a quantitative financial analyst. For each news snippet
494     about a stock you will output exactly four integer scores on a 1-5
495     scale, separated by a pipe '|', in this exact order: sentiment |
496     risk | confidence | volatility_forecast. (Per-axis rubrics and
497     canonical few-shot exemplars are lengthy; we ship the verbatim
498     templates with the scoring scripts in the public repository.) When
499     multiple news items are given for one batch, output one line per
500     item. Never add explanation -- only scores.
```

501 Two few-shot examples are prepended to each batch request. Up to 20 articles are batched per API
502 call to amortise latency.

503 A.2 Network Architecture

	Hyperparameter	Value
	Hidden layers	2×512 (Tanh)
	Observation dim	421
	Action dim	30
	PPO clip	0.2
504	Learning rate	3×10^{-4}
	Discount γ	0.99
	GAE λ	0.97
	Steps per epoch	$4,000 \times 4$ workers
	Train epochs	30
	Max KL	0.01
	Drawdown penalty α	0.1

505 A.3 Data Pipeline

506 Stock price data: Yahoo Finance via `yfinance` (adjusted closes, 2013-01-02–2023-12-29, 30
507 NASDAQ-100 tickers). Technical indicators computed via `stockstats`. Turbulence index com-
508 puted following Kritzman and Li [2010]. News data: Hugging Face dataset `Zihan1004/FNSPID`,
509 filtered to the 30 target tickers, 40,850 articles collected (2009–2023).

510 A.4 Compute

511 All experiments were run on a single MacBook Pro (Apple M-series, 16 GB unified memory). LLM
512 scoring: ≈ 17.5 hours for 40,850 articles at batch size 20 (OpenRouter API, `gpt-oss-20b:free`
513 free-tier model). DP-PPO training: ≈ 45 minutes per 30-epoch run (4 MPI workers, CPU-only
514 PyTorch).